Process book

perspectiva

https://github.com/com-480-data-visualization/perspectiva

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Introduction

Our website is based on GDELT data, GKG v2 (GDELT project, n.d), where we have studied news articles related to Elon Musk from 2015 to today, analysing sentiment over time between countries. This process book describes the process and iterations of ideas, some of the mathematical methods used to drive our design descisions, the design decisions, challenges, and delegation of work.

Delegation of work

Our ideation process began individually: each team member conducted their own exploratory data analysis on different datasets (vaccines, stock market and Social Media). We explored how specific events and topics were reported, covered, and framed differently across the world and over time, based on the GDELT dataset, which is an archive of online newspaper data, and AI sentiment analysis. Our goal was to find out how the same moments in history have been portrayed differently depending on location, and how these portrayals have changed through time.

We then came together as a group to share findings, discuss what we found interesting, and debate possible directions for the project. Through this discussion and iterations that will be explained in the next chapter of this process book, we pivoted the project to focus on having a more narrative storytelling website analysing news articles about Elon Musk. We chose to divided responsibilities based on each person's strengths.

Each team member took ownership of a core area while staying in dialogue with the others, by having group meetings, giving feedbacks to eachother, and communicating and discussing ideas and new directions as they emerged. No workstream operated in isolation, and decisions in one area informed and shaped the others.

A strength of our team was the diversity of our backgrounds: Andrine brought design expertise, Gaspar strong engineering and programming skills, and Vinayak experience in machine learning and data analysis. This complementarity meant each person could lead confidently in their domain while contributing meaningfully to the others.

Anoverview of each persons main area:

- Gaspar was responsible for building the website skeleton and developing the final implementation.
- Vinayak led the in-depth EDA and data processing. He also impleented visualisations to the final website.
- Andrine was responsible for the narrative structure and visual hierarchy, producing wireframes and a high-fidelity prototype in Figma, as well as having main responsibility for the process book.

Process

The initial idea came from Radio Garden's spatial browsing model. This is a globe-based interface where users can browse live radio from anywhere in the world. We wanted to apply the same concept to news, by having a world map where users could select a topic and observe how the coverage spread geographically, how volume changed over time, and how sentiment shifted by country. Alongside the map, we were planning to add supplementary 2D views with charts, graphs, and network diagrams to offer deeper analysis. This required a visualization architecture flexible enough to work across multiple topics.

As mentioned, the project draws on the GDELT Global Knowledge Graph v2 (GKG v2) (GDELT project, n.d), queried via Google BigQuery. GDELT monitors news media in over 100 languages in near real-time, tagging each article with geographic mentions, sentiment scores, and thematic classifications. We identified some data quality issues early: aggregator pollution from syndication outlets, location ambiguity in V2Locations, duplicate records, and query cost at full table scale. These were addressed each through filtering, deduplication on DocumentIdentifier, and partition-based querying.

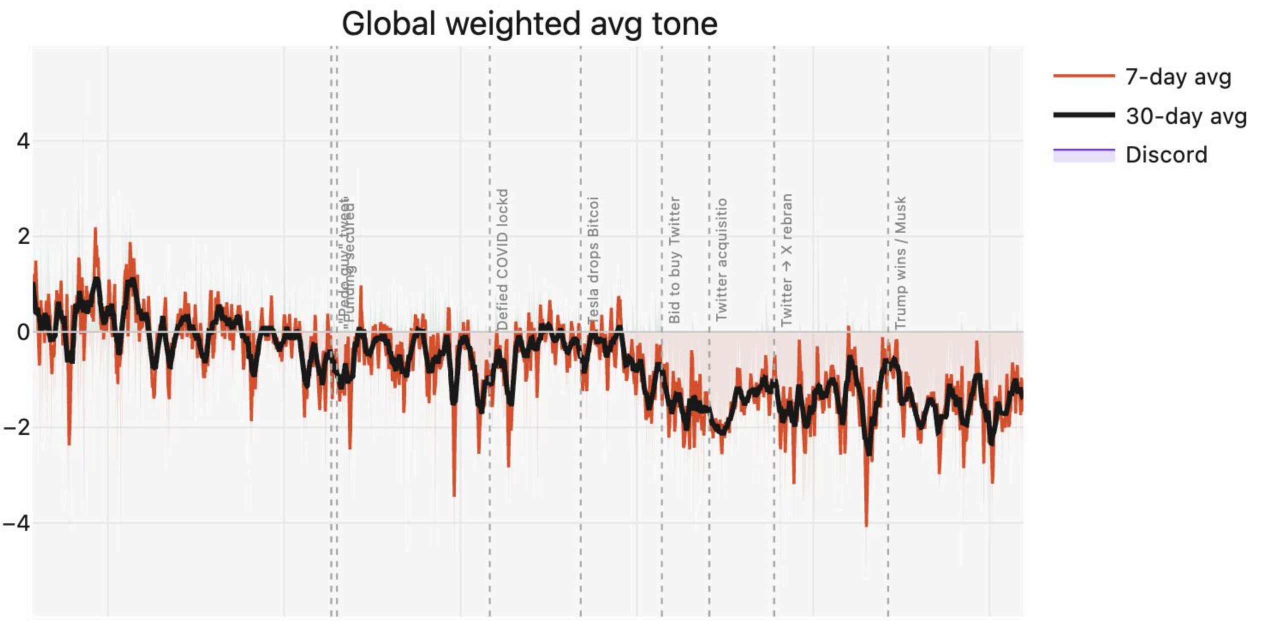
The initial topic set was vaccines, the stock market, and social media. These were chosen for their global relevance and the expectation that coverage patterns would differ meaningfully across countries. We chose to first focus only on Vaccines in relation to Covid 19, and then add other topics later to scale up. After early exploratory work with the vaccine dataset, it became clear that a more polarizing subject would produce more variation in both volume and sentiment. We wanted a topic where something had actually changed a lot over time, where countries had not agreed, and where that disagreement would be visible in the data.

We had to do another round of ideation to find topics that were more interesting to work on, and that would have a stronger storyline. Michael Jackson came up as a strong candidate. We used Claude Ai to get a quick overview of how his narrative story could be, before doing of own analysis. Michael Jackson is globally covered, tonally complex, and has a clear narrative arc; celebrated for decades, then accused, then defended, then contested again after his death (and he is still relevant and discussed today). The design approach shifted accordingly: rather than an open-ended explorer, we moved toward a more guided narrative structure where the map would set the context, and a sequenced story-driven visualization would carry the user through key moments and eras.

With this shift, we created new wireframes and an updated structural foundation, while still keeping news coverage and sentiment over time as the base. However, further analysis of the GDELT dataset revealed that the GKG v2 dataset we needed to use beacause it had rich tone data and was partitioned (much faster/cheaper to query), only began in February 2015. Earlier versions either lack tone data entirely (GDELT 1.0 Events tracks political and diplomatic events, not article sentiment), or use a less structured schema that didn’t match our pipeline (GKG v1, starting April 2013). Since only parts of Michael Jackson's story happened after 2015, the dataset could not support the narrative scope that the project required.

With the 2015 data floor as a constraint, we ran a second round of topic ideation. We used AI to help us with this ideation, by asking what news topics today that likely has and interesting arc in sentiment and volum. Together, we went through the different topics and found these topics to be more interesting:

- Sputnik
- Brexit
- Elon Musk



Elon Musk EDA: Global wighted agerage tone over time (from 2015 to 2026), showing a 7 and 30 day average.

We did some exploratory data analysis on the three topics, and concluded that Elon Musk's data stood out the most, as he had a very distinct shift in senitment over time. His news coverage also broke naturally into distinct periods with clear shifts in tone and volume. The math behind this can be found under "Further EDA: Elon Musk". The most striking finding was the overall downward trajectory of sentiment across the full dataset: from a mean above zero in 2015, where coverage was broadly positive, to below zero by 2026. That arc gave us a story worth telling, and it raised the questions we wanted the visualization to answer: which events drove the biggest shifts in how he is perceived? Which countries saw him as an innovator, and which were more negative? How have the words different countries use to describe him changed over time?

Further EDA: Elon musk

We analysed 921,348 GDELT articles across 157 countries (Feb 2015 – May 2026). The headline finding: global weighted average tone is –1.201, with only 26.3% of country-days registering positive sentiment. 2015 is the only net-positive year (+0.08); by 2026, tone collapsed to –1.76. The 15-chart EDA revealed three structural patterns. First, the global trajectory shows a monotonic decline (with only one recovery period during covid). Second, geopolitical blocs (G7, BRICS, Anglosphere, Global South) track each other in near-perfect lockstep, ruling out political or regional explanations. Third, coverage concentration is extreme: three Anglophone countries (USA, UK, Australia) produce 48% of all articles.

We used three quantitative methods to identify inflection points:

1. CUSUM (Cumulative Sum Control Chart): We computed CUSUM_t = Σ(daily_tone_i − global_mean) for all days. The CUSUM curve inflects when the underlying mean shifts. We found the first date where the 90-day rolling mean crossed –1.0 and held: March 2020. This is the statistically defensible structural break.
2. Event impact windows: For each candidate event, we extracted a ±30-day window of global tone. We computed impact = mean(tone[t:t+30]) − mean(tone[t-30:t]) — the difference between post-event and pre-event baseline. Events with the largest negative delta and longest recovery time (measured as days until tone returned to baseline, if ever) were selected. The Twitter acquisition (Oct 2022) never recovered.
3. Rolling Pearson correlation: We computed 90-day rolling r(volume, tone). The date this correlation turned persistently negative — meaning more coverage = worse coverage — marked when fame became a liability (early 2026).

Events were then selected to represent distinct types of controversy (personal, financial, institutional, political) with no two occupying the same narrative role, producing the final six: Pedo Guy → COVID defiance → Twitter acquisition → UK riots → Nazi salute → DOGE.

The sentiments score from GDELT and volume compared to country population size defined which countries had the best and the worst sentiment score.

We had some struggles finding out how to find meaningful words for the worldcloud, but figured that GDELT doesn’t provide actual topics it is mostly the names of the organization and the classification tags for individual news items, in-order to work with this, we used proper nouns and themes form the classification tags for our word-cloud.

The design

Martini Glass Structure

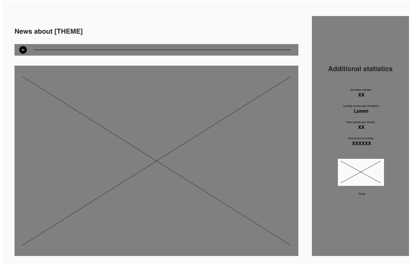
We planned on following the martini glass structure from the start, but the excecution of it changed. The first version placed the map and the narrative content side by side on the same screen. The idea was for users to could expand the storytelling panel by clicking on it. However, this made the storytelling passive. The user was in control of whether to engage with the narrative at all, and the author-driven part of the experience felt optional and very secondary to the exploratory part. We wanted to guide the reader more through the insights, show actively what was important, and build towards a conclusion.

This was part of the reason to shift from the multi-theme explorer to the Elon Musk narrative. We redesigned the structure: rather than placing exploration and storytelling side by side, the visualization now moves the user through a sequence of distinct views. The landing page establishes the theme and shows summarizing statistics with towards what they can expect from the website. The map then comes as the second view, functioning as an introduction to how sentiment toward Elon Musk has changed across the world over time, with the five eras visible as color shifts across countries. This view was designed to raise questions in the user: why are some countries more negative? What are these eras? What happened to cause those shifts? Those questions were supposed to guide the user into the next section, where each era is examined. Finally, the summary at the end answers the questions the user might have after the storytelling. The map is not the primary analytical tool anymore, but instead the entry point that draws the user into the story.

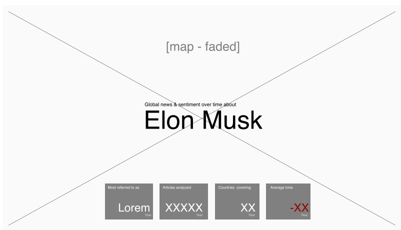
The Landing Page

The first thing a user sees is the landing page: the subject, and four summary statistics: most referred to as, articles analyzed, countries covering, and average tone. Before any interaction, the user gets an idea for what to expect from the website. The average tone is shown in red, already signaling that the overall story trends negative. The world map is visible behind the text, muted and non-interactive, setting geographic scope.

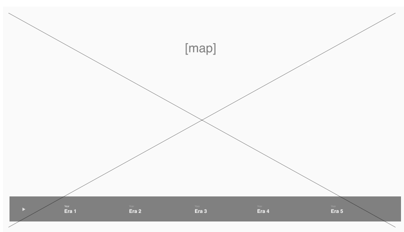
Initial idea



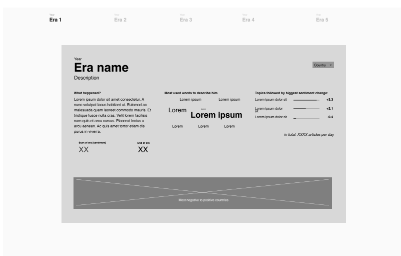
Landing page



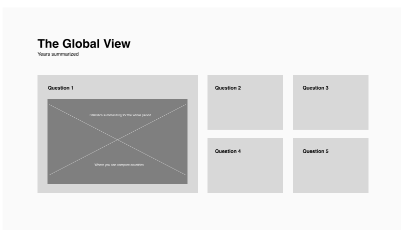
Map - play timeline



Eras - go through all 5



Summary - choose a question



The Sentiment Chart

The third view is the full sentiment timeline; a 45-day moving average line over the complete time period, with the five eras shown, and annotated key events that drove big shifts. Below the line chart, a volume bar chart shows article count. We chose to have this as a summary of the map, allowing the users to see in one picture how sentiment changed over time.

The Era Detail Pages

Prototype (real website was changed)

After the sentiment chart, the user is guided thorugh each era, with its own detailed page structured around three panels. On the left, there is a a short text description based on the news articles from that era. In the center, there si a word cloud showing the most frequently used words to describe Musk. On the right, there is a ranked list shows news topics that were followed by the largest sentiment changes. Below the three panels, there is a a ranked row of countries running from most negative to most positive. In the final prototype , we decided to remove the topics, and instead focus on the wordcloud, beacause it required less cognitive load by the user. Though wordclouds can be impresize, and users could struggle to differenciate between long words and bigger fonts, we chose to use it anyway. In this context, the goal is not precise measurement, but to communicate how the narrative shifted between the eras. What's more important than the exact size of the words, is the difference in the word usage.

The Global Summary

The final section pulls everything together. After moving through the five eras in sequence, the user arrives at a view that answers the questions the narrative might have raised: which countries changed their opinion the most, which were consistently the most negative and most positive across the whole period, and how the language used to describe Musk shifted globally from 2015 to 2026.

Color scale

Prototype (with placeholder text)

Current colors

Alternative colors

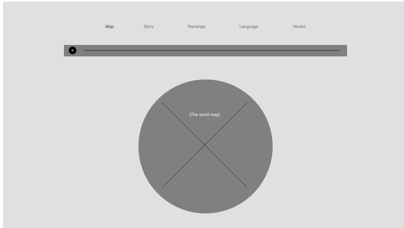
Sentiment is shown through numbers from negative to positive, and a colorscale from red(negative) to green(positive) and white/dark blue as neutral. The diverging color scale is the appropriate choice for this data type, combining two sequential palettes that meet at a neutral center. We chose the colors beacause of the intuitive meaning that most users will immediately understand. We started off having the two colors fade into eachother, but this introduced a third color that would confuse the user.

We acknowledge that an important critique of our design is that red-green is the most common form of colorblindness. We chose to prioritize the intuitive legibility of red-green for the majority of users. We also have a plus an minus in front of all numbers indicating sentiment. A future implementation could be to add a universal design alternative (eg a toggle bar changing to orange-purple scale).

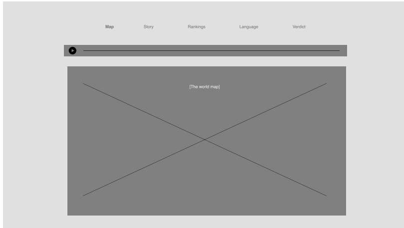
Globe vs. Flat Map

The earliest prototype used a 3D globe, inspired directly by Radio Garden's model (2019) of spinning through a spatial dataset. Visually, the globe was appealing and it matched the global scope of the data and gave the interface an immediate sense of scale. However, it created a problem: only half the world is visible at any given moment. To compare how different countries covered the same story, a user had to rotate the globe, hold information in memory, and mentally re-orient with each turn, while at the same time, interesting data might show on the other side of the globe. The core goal about comparing sentiment across countries over time did not work with this method.

LoFi wireframes of globe page



LoFi wireframes of flat map page



Therefore, we changed to the flat map. All countries are visible at the same time, and the user can scan across the full dataset with less effort. This reduced the cognitive work required to extract meaning. A globe is a compelling object, but for country-level comparison over time, a flat projection is simply more user friendly.

Temporal Aggregation: From Daily to Mean

The first version of the map updated sentiment on a day-to-day basis. This turned out to be difficult to read. Sentiment in the GDELT data is noisy, and a single day's coverage can spike sharply in either direction based on one or two major headlines, then return to baseline the next day. To follow how a country's sentiment was shifting over time, a user would need to remember what color it was "yesterday", compare it to "today", and decide whether the change was meaningful or just noise. That is a large amount of cognitive work to place on the user for 154 countries at the same time, and it works against the goal of making trends visible.

The exact amount of days were iterated on, but we concluded that switching to a 45, 90 and and 300 days rolling mean for the map and a 45 mean for the graph under it, resolved this the best. When a country changes color, the change is meaningful and the user could see the chagne from one era to the next with less cognitive load. For the map, we also added a mean score for the whole world. A critique of our final design is that this section still feels somewhat visually noisy. However, the third page with the sentiment chart for the whole period summarizes this and gives the user an overview of what they just explored.

Dark mode and font

The dark blue mode and sans-serif typography were chosen to match the subject: Elon Musk is a technology figure, and the aesthetics fits the data with sentiment trending negative across the full period. A light design would feel too lightweight and cheerful for the story being told. The neutral blue color was also chosen beacause we wanted the design to feel neutral, by not expressing any opinion of our own, and instead let the news and analysis speak for the sentiment.

Use of AI

As English is not our first language, we used Claude AI was to refine the language in this process book, to shorten down text and make it more concise. During ideation phase, AI assisted in suggesting potential themes and directions. While the main structure and all wireframes from low to high fidelity prototypes and designs in Figma were created by us, AI was used to critique and evaluate design decisions along the way.

We used AI throughout our development process in three key ways: first, to accelerate repetitive tasks; second, to maintain clean, consistent code formatting across our components; and third, to traverse and debug our codebase, catching errors and ensuring the visualizations rendered correctly with real data. It was in no-way helping us write our code, but was useful in catching errors and make sure the working across codebases is smooth.

Resources

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Tools:

- Next.js (React) & Tailwind CSS
- MapLibre GL & react-map-gl
- Google BigQuery
- Python (Pandas & PyArrow)
- Github Pages & Actions
- Figma
- Claude AI